Linhai Song

Professor
Institute of Computing Technology
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RESEARCH INTERESTS

Improving the reliability, security and efficiency of AI systems and software systems

EDUCATION

University of Wisconsin–Madison, Madison, WI, USA — Nov. 2015
Ph.D., Computer Science (M.S. along the way)
Advisor: Shan Lu

Chinese Academy of Sciences, Beijing, China — Jun. 2010
M.S., Computer Science

Huazhong University of Science and Technology, Wuhan, Hubei, China — Jun. 2007
B.E., Software Engineering

EMPLOYMENT

Institute of Computing Technology, Beijing, China — Aug. 2025 - Now
Professor

Huawei Technologies Co., Ltd, Beijing, China — Nov. 2022 - July 2025

Pennsylvania State University, State College, PA, USA — May 2023 - June 2024
Associate Professor at the College of Information Sciences and Technology

Pennsylvania State University, State College, PA, USA — Aug. 2017 - May 2023
Assistant Professor at the College of Information Sciences and Technology

Kwai Inc., Seattle, WA, USA — Jun. 2021 - Aug. 2021
Consultant

ByteDance Ltd., Palo Alto, CA, USA — May 2019 - Aug. 2019
Consultant

FireEye, Inc., Milpitas, CA, USA — Nov. 2015 - Jul. 2017
Staff Research Scientist

NEC Laboratories America, Inc., Princeton, NJ, USA — May 2013 - Aug. 2013
Research Intern

Microsoft Research Asia, Beijing, China — May 2010 - Jul. 2010
Research Intern

HONORS AND AWARDS

- **NSF CAREER Award**, 2022
- **Mozilla Research Award**, 2019
- **MICRO'2014 Best Paper Runner Up** for paper [C5], 2014
- **ACM SIGPLAN Research Highlights @ PLDI** for paper [C1], 2011

PUBLICATIONS[1]

Refereed Journal Articles

[J4] Mengting He[S], Shihao Xia[S], Boqin Qin[S], Nobuko Yoshida, Tingting Yu, Yiying Zhang, and **Linhai Song**. "How to Save My Gas Fees: Understanding and Detecting Real-World Gas Issues in Solidity Programs." In *Transactions on Software Engineering (**TSE**)*, 2025. ***Accepted as a Journal-First paper by ICSE'2026***

[J3] Boqin Qin[S], Yilun Chen, Haopeng Liu, Hua Zhang, Qiaoyan Wen, **Linhai Song**, and Yiying Zhang. "Understanding and Detecting Real-World Safety Issues in Rust." In *Transactions on Software Engineering (**TSE**)*, 2024.

[J2] Boqin Qin[S], Tengfei Tu[S], Ziheng Liu[S], Tingting Yu, and **Linhai Song**. "Algorithmic Profiling for Real-World Complexity Problems." In *Transactions on Software Engineering (**TSE**)*, 2021. ***Accepted as a Journal-First paper by ICSE'2022***

[J1] Dongdong Deng, Guoliang Jin, Marc de Kruijf, Ang Li, Ben Liblit, Shan Lu, Shanxiang Qi, Jinglei Ren, Karthikeyan Sankaralingam, **Linhai Song**, Yongwei Wu, Mingxing Zhang, Wei Zhang, and Weimin Zheng. "Fixing, Preventing, and Recovering from Concurrency Bugs." In *Science China Information Sciences volume*, vol. 58, pp. 1–18, April 2014.

Refereed Conference Proceedings

[C20] Mengting He[S], Shihao Xia[S], Haomin Jia[S], Wenfei Wu, and **Linhai Song**. "M2K: Making the Model-Kernel Interface Explicit for Reliable CUDA Kernel Verification." In *Proceedings of the 32nd ACM Symposium on Operating Systems Principles (**SOSP'2026**)*, Sep 2026.

[C19] Shihao Xia[S], Mengting He[S], Shuai Shao, Tingting Yu, Yiying Zhang, Nobuko Yoshida, and **Linhai Song**. "SymGPT: Auditing Smart Contracts via Combining Symbolic Execution with Large Language Models." In *Proceedings of the 2026 ACM International Conference on Object Oriented Programming Systems Languages & Applications (**OOPSLA'2026**)*, Oct 2026.

[C18] Wenzhang Yang, **Linhai Song**, and Yinxing Xue. "Rust-lancet: Automated Ownership-Rule-Violation Fixing with Behavior Preservation." In *Proceedings of the 46th International Conference on Software Engineering (**ICSE'2024**)*, April 2024.

[C17] Stephen Ellis*, Shuofei Zhu*[S], Nobuko Yoshida, and **Linhai Song**. "Generic Go to Go: Dictionary-Passing, Monomorphisation, and Hybrid." In *Proceedings of the 2022 ACM International Conference on Object Oriented Programming Systems Languages & Applications (**OOPSLA'2022**)*, Dec 2022. (Acceptance Rate: 31.3%, 92 out of 294) (*: co-first authors)

[C16] Shuofei Zhu*[S], Ziyi Zhang*[S], Boqin Qin[S], Aiping Xiong, and **Linhai Song**. "Learning and Programming Challenges of Rust: A Mixed-Methods Study." In *Proceedings of the 44th International Conference on Software Engineering (**ICSE'2022**)*, May 2022. (Acceptance Rate: 28.5%, 197 out of 691) (*: co-first authors)

[1]Students directly under my supervision are denoted by "S".

[C15] Ziyi Zhang[S], Shuofei Zhu[S], Jaron Mink, Aiping Xiong, **Linhai Song**, and Gang Wang. "Beyond Bot Detection: Combating Fraudulent Online Survey Takers." In *Proceedings of the ACM Web Conference 2022 (**WWW'2022**)*, April 2022. (Acceptance Rate: 17.7%, 323 out of 1822)

[C14] Ziheng Liu*[S], Shihao Xia*[S], Yu Liang, **Linhai Song**, and Hong Hu. "Who Goes First? Detecting Go Concurrency Bugs via Message Reordering." In *Proceedings of the 27th International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS'2022**)*, March 2022. (Acceptance Rate: 20.1%, 80 out of 397) (*: co-first authors)

[C13] Ziheng Liu[S], Shuofei Zhu[S], Boqin Qin[S], Hao Chen, and **Linhai Song**. "Automatically Detecting and Fixing Concurrency Bugs in Go Software Systems." In *Proceedings of the 26th International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS'2021**)*, April 2021. (Acceptance Rate: 18.8%, 75 out of 398)

[C12] Boqin Qin[S]*, Yilun Chen*, Zeming Yu[S], **Linhai Song**, and Yiying Zhang. "Understanding Memory and Thread Safety Practices and Issues in Real-World Rust Programs." In *Proceedings of the 41st ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI'2020**)*, June 2020. (Acceptance Rate: 22.5%, 77 out of 341) (*: co-first authors)

[C11] Shuofei Zhu[S], Jianjun Shi[S], Limin Yang, Boqin Qin[S], Ziyi Zhang[S], **Linhai Song**, and Gang Wang. "Measuring and Modeling the Label Dynamics of Online Anti-Malware Engines." In *Proceedings of the 29th USENIX Security Symposium (**USENIX Security'2020**)*, August 2020. (Acceptance Rate: 17.1%, 44 out of 256)

[C10] Bangwen Deng, Wenfei Wu, and **Linhai Song**. "NFReducer: Redundant Logic Elimination in Network Functions." In *Proceedings of the 2020 ACM SIGCOMM Symposium on SDN Research (**SOSR'2020**)*, March 2020. (Acceptance Rate: 28.3%, 17 out of 60)

[C9] Peng Peng, Limin Yang, **Linhai Song**, and Gang Wang. "Opening the Blackbox of VirusTotal: Analyzing Online Phishing Scan Engines." In *Proceedings of the 2019 ACM Internet Measurement Conference (**IMC'2019**)*, October 2019. (Acceptance Rate: 19.7%, 39 out of 197)

[C8] Tengfei Tu[S], Xiaoyu Liu, **Linhai Song**, and Yiying Zhang. "Understanding Real-World Concurrency Bugs in Go." In *Proceedings of the 24th International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS'2019**)*, April 2019. (Acceptance Rate: 21.1%, 74 out of 350)
The second-most visited URL related to Golang in 2019. Featured on "a morning paper" and "Hacker News".

[C7] **Linhai Song** and Shan Lu. "Program Analysis for Inefficient Loops." In *Proceedings of the 39th International Conference on Software Engineering (**ICSE'2017**)*, May 2017. (Acceptance Rate: 16.4%, 68 out of 415)

[C6] Rui Gu, Guoliang Jin, **Linhai Song**, Linjie Zhu, and Shan Lu. "What Change History Tells Us About Thread Synchronization." In *Proceedings of the 2015 10th Joint Meeting on Foundations of Software Engineering (**FSE'2015**)*, August 2015. (Acceptance Rate: 25.4%, 74 out of 291)

[C5] **Linhai Song**, Min Feng, Nishkam Ravi, Yi Yang, and Srimat Chakradhar. "COMP: Compiler Optimizations for Manycore Processors." In *Proceedings of the 47th Annual IEEE/ACM International Symposium on Microarchitecture (**MICRO'2014**)*, December 2014. (Acceptance Rate: 19.4%, 53 out of 273)
MICRO'2014 Best Paper Runner Up.

[C4] **Linhai Song** and Shan Lu. "Statistical Debugging for Real-World Performance Problems." In *Proceedings of the 2014 ACM International Conference on Object Oriented Programming Systems Languages & Applications (**OOPSLA'2014**)*, October 2014. (Acceptance Rate: 28.4%, 53 out of 186)

[C3] Adrian Nistor, **Linhai Song**, Darko Marinov, and Shan Lu. "Toddler: Detecting Performance Problems via Similar Memory-Access Patterns." In *Proceedings of the 2013 International Conference on Software Engineering (**ICSE'2013**)*, May, 2013. (Acceptance Rate: 18.5%, 85 out of 461)

[C2] Guoliang Jin*, **Linhai Song***, Xiaoming Shi, Joel Scherpelz, and Shan Lu. "Understanding and Detecting Real-World Performance Bugs." In *Proceedings of the 33rd ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI'2012**)*, June 2012. (Acceptance Rate: 18.8%, 48 out of 255) (*: co-first authors)

[C1] Guoliang Jin, **Linhai Song**, Wei Zhang, Shan Lu, and Ben Liblit. "Automated Atomicity-Violation Fixing." In *Proceedings of the 32nd ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI'2011**)*, June 2011. (Acceptance Rate: 23.3%, 55 out of 236)
***ACM SIGPLAN Research Highlights Award** (Top 8 papers selected from all papers in 13 SIGPLAN conferences in 2011 for "high quality and broad appeal").*

Refereed Workshop Proceedings

[W5] Xiheng Li[S], Mengting He[S], Chengcheng Wan, and **Linhai Song**. "Mobile-MCP: Implementing the Model Context Protocol via Android's Inter-Application Communication Mechanisms." In *Proceedings of the 1st Workshop on Operating Systems Design for AI Agents (**AgenticOS'2026**)*, March 2026.

[W4] Shihao Xia[S], Mengting He[S], **Linhai Song**, and Yiying Zhang. "SC-Bench: A Large-Scale Dataset for Smart Contract Auditing." In *Proceedings of the 2nd International Workshop on Large Language Models for Code (**LLM4Code'2025**)*, May 2025.

[W3] Yongheng Chen[S], **Linhai Song**, Xinyu Xing, Fengyuan Xu, and Wenfei Wu. "Automated Finite State Machine Extraction." In *Proceedings of the 3rd ACM Workshop on Forming an Ecosystem Around Software Transformation (**FEAST'2019**)*, November 2019. (Acceptance Rate: 87.5%, 7 out of 8)

[W2] **Linhai Song** and Xinyu Xing. "Fine-Grained Library Customization." In *Proceedings of the First International Workshop on SoftwAre debLoating And Delayering (**SALAD'2018**)*, July 2018. (Acceptance Rate: 66.7%, 2 out of 3)

[W1] **Linhai Song**, Heqing Huang, Wu Zhou, Wenfei Wu, and Yiying Zhang. "Learning from Big Malware." In *Proceedings of the 7th ACM SIGOPS Asia-Pacific Workshop on Systems (**APSys'2016**)*, August 2016. (Acceptance Rate: 40.8%, 20 out of 49)

Technical Reports

[T10] Shihao Xia[S], Mengting He[S], Haomin Jia[S], and **Linhai Song**. "Doc2Spec: Synthesizing Formal Programming Specifications from Natural Language via Grammar Induction." arXiv:2602.04892.

[T9] Shihao Xia[S], Mengting He[S], Shuai Shao, Tingting Yu, Yiying Zhang, Nobuko Yoshida, and **Linhai Song**. "SymGPT: Auditing Smart Contracts via Combining Symbolic Execution with Large Language Models." arXiv:2502.07644.

[T8] Chenqi Zhao, Wenfei Wu, **Linhai Song**, and Yuchen Xu "MicroMoE: Fine-Grained Load Balancing for Mixture-of-Experts with Token Scheduling." arXiv:2511.16947.

[T7] Shihao Xia[S], Mengting He[S], **Linhai Song**, and Yiying Zhang. "SC-Bench: A Large-Scale Dataset for Smart Contract Auditing." arXiv:2410.06176.

[T6] Shihao Xia[S], Shuai Shao, Mengting He[S], Tingting Yu, **Linhai Song**, and Yiying Zhang. "AuditGPT: Auditing Smart Contracts with ChatGPT." arXiv:2404.04306.

[T5] Mengting He[S], Shihao Xia[S], Boqin Qin[S], Nobuko Yoshida, Tingting Yu, **Linhai Song**, and Yiying Zhang. "How to Save My Gas Fees: Understanding and Detecting Real-world Gas Issues in Solidity Programs." arXiv:2403.02661.

[T4] Zeming Yu[S], **Linhai Song**, and Yiying Zhang. "Fearless Concurrency? Understanding Concurrent Programming Safety in Real-World Rust Software." arXiv:1902.01906.

[T3] **Linhai Song** and Xinyu Xing. "Fine-Grained Library Customization." arXiv:1810.11128.

[T2] **Linhai Song** and Shan Lu. "Program Analysis for Inefficient Loops." UChicago CS Technical Report TR-2016-06.

[T1] **Linhai Song** and Shan Lu. "Statistical Debugging for Real-World Performance Problems." UW-Madison CS Technical Report 1803.

Posters

[P3] Ziyi Zhang[S] and **Linhai Song**. "Poster: Visualizing Critical Sections in Rust." In *Student Research Competition at the 27th ACM Symposium on Operating Systems Principles (**SOSP'2019**)*.

[P2] Tengfei Tu[S], Xiaoyu Liu, **Linhai Song**, and Yiying Zhang. "Poster: Understanding Real-World Concurrency Bugs in Go." In *the 13rd USENIX Symposium on Operating Systems Design and Implementation (**OSDI'2018**)*.

[P1] **Linhai Song** and Shan Lu. "Poster: Statistical Debugging for Real-World Performance Problems." In *the 4th Greater Chicago Area Systems Research Workshop (**GCASR'2015**)*.

Demonstrations

[D2] Ziyi Zhang[S], Boqin Qin[S], and **Linhai Song**. "Demo: VRLifeTime -- An IDE Tool to Avoid Concurrency and Memory Bugs in Rust." In *the 27th ACM Conference on Computer and Communications Security (**CCS'2020**)*.

[D1] Shuofei Zhu[S], Ziyi Zhang[S], Limin Yang, **Linhai Song**, and Gang Wang. "Demo: Benchmarking Label Dynamics of VirusTotal Engines." In *the 27th ACM Conference on Computer and Communications Security (**CCS'2020**)*.

Software and Data Release

[S10] A framework and protocol for LLMs to communicate with mobile APPs, 2026.
https://github.com/system-pclub/mobile-mcp.

[S9] An LLM-powered verification tool for Solidity smart contracts, 2026.
https://github.com/symbolic-gpt/symgpt.

[S8] A large-scale dataset for smart contract auditing, 2025.
https://github.com/charlesxsh/scbench.

[S7] A static Rust memory and concurrency bug detector, 2023.
https://github.com/BurtonQin/lockbud. (***587*** GitHub stars)

[S6] A dynamic Go concurrency bug detector, 2022.
https://github.com/system-pclub/GFuzz. (***137*** GitHub stars)

[S5] A static Go concurrency bug detector, 2021.
https://github.com/system-pclub/GCatch. (***444*** GitHub stars)

[S4] A production-run algorithmic profiler, 2021.
https://github.com/ComAirProject/ComAir.

[S3] Dataset of the daily snapshots of VirusTotal labels for 14,000 files over a year, 2020.
https://sfzhu93.github.io/projects/vt/index.html.

[S2] Dataset of 170 real-world Rust safety issues, 2020.
https://github.com/system-pclub/rust-study. (***100*** GitHub stars)

[S1] Dataset of 171 real-world Go concurrency bugs, 2019.
https://github.com/system-pclub/go-concurrency-bugs. (***260*** GitHub stars)

Patents

[PA1] Min Feng, Srimat Chakradhar, and **Linhai Song**. "Compiler Optimization for Many Integrated Core Processors." U.S. Patent No. 20150277877, October 1st, 2015.

PROFESSIONAL ACTIVITIES

Conference Program Committee Service

- ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA**): 2026
- USENIX Annual Technical Conference (**ATC**): 2024, 2025, 2026
- ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (**PPoPP**): 2024, 2027
- European Conference on Computer Systems (**EuroSys**): 2024
- International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**): 2022, 2026
- International Workshop on Large Language Models for Code (**LLM4Code**): 2026
- Workshop on Programming Languages and Operating Systems (**PLOS**): 2021, 2023, 2025
- Poster and Demonstration Session at ACM Conference on Computer and Communications Security (**CCS**): 2020
- Poster Session at International Conference on Software Engineering (**ICSE**): 2020
- Software Engineering in Practice at International Conference on Software Engineering (**ICSE**): 2019
- ACM SIGOPS Asia-Pacific Workshop on Systems (**APSys**): 2018, 2019, 2022, 2023
- Student Research Competition (**SRC**) at ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**FSE**): 2018
- Student Research Competition (**SRC**) at International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**): 2018
- Artifact Evaluation at ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI**): 2015
- Artifact Evaluation at ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA**): 2014

Conference Reviewer

- International Symposium on Code Generation and Optimization (**CGO**): 2024
- ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA**): 2018
- International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**): 2019, 2020, 2021
- ACM Conference on Computer and Communications Security (**CCS**): 2017, 2018
- USENIX Annual Technical Conference (**USENIX ATC**): 2017

Journal Reviewer

- ACM Computing Surveys
- ACM Transactions on Computer Systems
- Empirical Software Engineering Journal
- IEEE Computer Architecture Letters
- Transactions on Software Engineering
- Journal of Computer Science and Technology
- IEEE/ACM Transactions on Networking
- Automated Software Engineering
- ACM Transactions on Software Engineering and Methodology

Journal Editor

- EAI Transactions on Security and Safety

Conference & Workshop Organization Service

- Chair for Student Research Competition (**SRC**) at International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**): 2019

Other Services

- Hong Kong Research Grants Council (**RGC**) ad-hoc reviewer: 2023
- Dutch Research Council (**NWO**) ad-hoc reviewer: 2023, 2024
- National Science Foundation (**NSF**) ad-hoc reviewer: 2021
- National Science Foundation (**NSF**) Review Panel: 2018

TALKS

- Understanding and Detecting Real-World Gas Issues in Solidity Programs
  - ICSE'2026, April 2026
- LLM-Assisted Software Verification
  - ChinaSoft, November 2025
- Understanding and Improving the Programmability of Rust
  - Peking University, April 2024
- Learning and Programming Challenges of Rust: A Mixed-Methods Study
  - Peking University, April 2023
- Detecting Concurrency Bugs in Go Software Systems
  - Yale University, June 2022
  - University of California, Santa Barbara, May 2022
  - University of Virginia, May 2022
  - Google, April 2022
  - University of Illinois Urbana-Champaign, April 2022
  - University of California, Merced, April 2022
  - Northwestern University, April 2022
  - Peking University, April 2022
- Who Goes First? Detecting Go Concurrency Bugs via Message Reordering

  - ASPLOS'2022, March 2022

- Combating Real-World Concurrency Bugs in Go
  - Bytedance, August 2021
  - Kwai, June 2021
  - Tsinghua University, May 2021
  - North Carolina State University, April 2021

- Measuring and Modeling the Label Dynamics of Online Anti-Malware Engines
  - Southern University of Science and Technology, January 2021
  - University of Science and Technology of China, December 2020

- Understanding Real-World Concurrency Bugs in Go
  - ASPLOS'2019, April 2019

- Understanding Real-World Concurrency Bugs in New Programming Languages
  - Carnegie Mellon University, October 2019
  - ByteDance, December 2018
  - Baidu X-lab, December 2018

- Fine-grained Library Customization
  - Salad'2018, July 2018

- Protocol Subsetting and Dialect Generation
  - Baidu X-lab, December 2017

- Performance Diagnosis for Inefficient Loops
  - ICSE'2017, May 2016

- Improve Software Security and Performance through Data Analytics
  - Pennsylvania State University, March 2016

- Learning from Big Malware
  - Baidu X-lab, May 2017
  - APSys'2016, August 2016

- Understanding, Detecting, and Diagnosing Real-World Performance Bugs
  - National University of Singapore, March 2016
  - Microsoft Research Asia, December 2015
  - Peking University, June 2015
  - Pivotal Labs, May 2015

- Statistical Debugging for Real-World Performance Problems
  - OOPSLA'2014, October 2014
  - WISDOM Workshop II, May 2014

- Optimizing Memory Performance on Many Integrated Core Coprocessors
  - NEC Labs America, August 2013

- Understanding and Detecting Real-World Performance Bugs
  - PLDI'2012, June 2012
  - Programming Languages Seminar, University of Wisconsin-Madison, May 2012

TALK VIDEOS

- "Mobile-MCP: Implementing the Model Context Protocol via Android's Inter-Application Communication Mechanisms" presented by Mengting He and Xiheng Li
  - https://www.youtube.com/watch?v=5648xuNEDTI (31 views)
  - https://www.bilibili.com/video/BV1qHX4BqEuN (634 views)
- "Algorithmic Profiling for Real-World Complexity Problems" presented by Boqin Qin
  - https://www.youtube.com/watch?v=M1hfaHPB868 (55 views)
  - https://www.bilibili.com/video/BV15Y4y187VQ (132 views)
- "Learning and Programming Challenges of Rust: A Mixed-Methods Study" presented by Shuofei Zhu
  - https://www.youtube.com/watch?v=STjQxTu3tS8 (***1157*** views)
  - https://www.bilibili.com/video/BV1RS4y1h7Nf (***1320*** views)
- "Who Goes First? Detecting Go Concurrency Bugs via Message Reordering" presented by Shihao Xia
  - https://www.youtube.com/watch?v=sQbnzYPOcz4 (109 views)
  - https://www.bilibili.com/video/BV1TF411b7nN (***4251*** views)
  - https://www.bilibili.com/video/BV1om4y1d7Jn (lightning talk) (***1512*** views)
- "Automatically Detecting and Fixing Concurrency Bugs in Go Software Systems" presented by Ziheng Liu
  - https://www.youtube.com/watch?v=qsjptcAWTWM (149 views)
  - https://www.bilibili.com/video/BV1Gb4y1D7QH (240 views)
- "Measuring and Modeling the Label Dynamics of Online Anti-Malware Engines" presented by Shuofei Zhu
  - https://www.youtube.com/watch?v=dhbcWx3HC64 (228 views)
  - https://www.bilibili.com/video/BV1Rk4y127cQ (728 views)
- "Understanding Memory and Thread Safety Practices and Issues in Real-World Rust Programs" presented by Yilun Chen
  - https://www.youtube.com/watch?v=s5UqjOEaZ_8 (542 views)
  - https://www.bilibili.com/video/BV17i4y1x7CM (***1904*** views)
  - https://www.bilibili.com/video/BV1zA411t7ze (lightning talk) (164 views)
- "Understanding Real-World Concurrency Bugs in Go" presented by Xiaoyu Liu
  - https://www.youtube.com/watch?v=ClVrJcTM-lA (388 views)
  - https://www.bilibili.com/video/BV12b411b7Gt (365 views)

GRANTS

- CAREER: Rethinking Toolchain Design for Rust
  - Role: PI;
  - Total: $550,193;
  - National Science Foundation (NSF);
  - 02/01/2022 to 01/31/2027.
- Avoiding Rust Deadlocks via Lifetime Visualization
  - Role: PI; with Yiying Zhang from UC San Diego as Co-PI;

  - Total: $60, 000; Personal Share: $30, 000 (50%);
  - Web3 Foundation;
  - 09/01/2021 to 08/31/2022.
- GCatch++: Automatically Detecting Concurrency Bugs in Software Systems implemented in Go
  - Role: PI;
  - Total: $30, 000;
  - Ethereum Foundation;
  - 09/01/2021 to 08/31/2022.
- Learning and Programming Challenges of Rust: An Interdisciplinary Investigation
  - Role: PI; with Aiping Xiong from Penn State as Co-PI;
  - Total: $55, 000; Personal Share: $27, 500 (50%);
  - IST@PSU Seed Grant;
  - 09/01/2021 to 08/31/2022.
- SaTC: CORE: Small: Understanding and Detecting Memory Bugs in Rust
  - Role: PI; with Hao Chen from UC Davis as Co-PI;
  - Total: $497,340; Personal Share: $298,404 (60%);
  - National Science Foundation (NSF);
  - 07/01/2020 to 12/31/2025.
- Measuring and Modeling the Label Dynamics of Online Anti-Malware Engines
  - Role: Sole PI;
  - Total: $9,966; Personal Share: $9,966 (100%);
  - ICDS@PSU Seed Grant;
  - 05/01/2020 to 04/30/2021.
- Statically Detecting Memory Bugs in Rust Applications
  - Role: Sole PI;
  - Total: $80,100; Personal Share: $80,100 (100%);
  - Open Tech Fund;
  - 01/01/2020 to 06/30/2021.
- Benchmarking Generic Functions in Rust
  - Role: Sole PI;
  - Total: $25,000; Personal Share: $25,000 (100%);
  - Mozilla Research Award;
  - 09/01/2019 to 08/31/2020.
- Benchmarking, Detecting, and Diagnosing Real-World Performance Problems
  - Role: Sole PI;
  - Total: $85,500; Personal Share: $85,500 (100%);
  - IST@PSU Seed Grant;
  - 09/01/2018 to 08/31/2019.

SELECTED PRESS

- **[Spectrum]** Scammers threaten quality of research survey data, 08/2023
- **[Software Engineering Daily]** Rust and Go Research with Linhai Song, 01/2021
- **[A Journey With Go]** Go: Concurrency Bugs in Go, 09/2019

- **[GTech Booster]** Is Rust the low-level-ish, 02/2019

TEACHING

Term	Course	Enrollment	Course Quality	Instructor Quality
Spring 2022	IST 597 Advanced Software Testing	5	7/7	7/7
FALL 2021	SRA 221 Information Security (1)	59	6/7	6/7
Spring 2021	SRA 221 Information Security (1)	60	6.5/7	6.5/7
Fall 2020	SRA 221 Information Security (1)	68	6/7	6/7
Fall 2019	IST 451 Network Security (1)	72	4.78/7	5.03/7
Fall 2019	IST 451 Network Security (2)	66	5.23/7	5.57/7
Fall 2018	IST 451 Network Security (1)	71	5.69/7	5.66/7
Fall 2018	IST 451 Network Security (2)	45	5.59/7	5.59/7
Spring 2018	IST 451 Network Security (1)	48	5.68/7	5.8/7
Fall 2017	IST 451 Network Security (1)	71	5.16/7	5.19/7

- New Courses Developed and Taught: IST 597
- Course Committee Members: SRA 221, Cyber 362
- Course Committee Chairs: IST 451
- Course Innovations:
  - Designed in-class quizzes, a weekly learning survey, a final exam, and eight new labs for IST 451;
  - Designed in-class quizzes, a weekly learning survey, two mid-term exams, and two new labs for SRA 221.

INTERNAL SERVICES

- Qualifying Exam Committee, Penn State, 2019, 2021, 2022, 2023, 2024
- Faculty Annual Review Committe, Penn State, 2022
- Faculty Search Committe, Penn State, 2020–2021
- Faculty Council, Penn State, 2019–2021
- Graduate Advisory Committee (GAC), Penn State, 2018–2019
- Graduate Recruiting Committee (GRC), Penn State, 2018–2020

ADVISING

Ph.D. Students

- Shihao Xia (co-advised with Hong Hu) (2021 – Present): [J4] [C20] [C19] [C14] [W4]
- Mengting He (2022 – Present) [J4] [C20] [C19] [W5] [W4]
- Haomin Jia (2025 – Present) [C20]
- Xiheng Li (2026 – Present) [W5]
- Shuofei Zhu (2018 – 2024): [C17] [C16] [C15] [C13] [C11] → Dr. Peng Liu's group
- Ziheng Liu (2018 – 2021): [J2] [C14] [C13] → Ph.D. at UCSD
- Li Wang (2021): → Assistant Professor at Fontbonne University

Master Students

- Xinyan Zhao (2026 – Present)
- Lenan Yu (2026 – Present)
- Tiffany Amigon (2022 – 2023)
- Shiqin Chen (2018 – 2021)

Undergraduate Students

- Edward Burke (2021)
- Erin Flannery (2020)
- Phil Reeves (2020)
- Tianchen Zhang (2019)

Visiting Students

- Boqin Qin (Ph.D. student from BUPT) (2018 – 2020): [J2] [J3] [C11] [C12] [C13] [C16] → China Telecom Cloud Computing
- Ziyi Zhang (Undergraduate from USTC) (2019 – 2021): [C11] [C15] [C16] → Ph.D. at Wisconsin-Madison
- Jianjun Shi (Ph.D. student from BIT) (2018 – 2019): [C11]
- Zeming Yu (2018 – 2019): [C12]
- Yongheng Chen (Undergraduate from NJU) (2019): [W3] → Ph.D. at Gatech
- Tengfei Tu (Ph.D. student from BUPT) (2017 – 2018): [J2] [C8] → faculty at BUPT

Thesis Committee at Penn State

- **[2025]** Siddhartha Balakrishna (PhD), Yu Liang (PhD).
- **[2024]** Lexiang Huang (PhD), Shuofei Zhu (PhD), Alexandar Devic (PhD), Zhixuan Huan (PhD).
- **[2022]** Saambhavi Baskaran (PhD).
- **[2022]** Minli Liao (PhD).
- **[2021]** Li Wang (PhD), Zhenpeng Lin (MS), Xian Wu (MS).

Qualification Committee at Penn State

- **[2024]** Yurui Chang, Zehao Liu, Zongyu Wu, Yaopei Zeng, Naixin Zhang.
- **[2023]** Song Liu, Yujia Wang, Hengkai Ye, Ziyi Yin.
- **[2022]** Zhimeng Guo, Geng Xiao, Junjie Xu, Huaisheng Zhu.
- **[2021]** Quan Li, Haizhou Wang, Tianrou Xia, Zhaohan Xi.
- **[2019]** Neisarg Dave, Ankur Mali, Shaurya Rohatgi, Rui Yu.

Student Mentoring Programs at Top-Tier Conferences

- **[ASPLOS'2022]** Xiang Cheng, Yuheng Yang.
- **[SOSP'2021]** Yuhan Liu, Hannah Atmer.
- **[ASPLOS'2021]** Shijia Wei , Jules Drean.